Phishing detection project documentation

Project Name:-

Phishing URL Detection System

Objective:-

To detect phishing websites by analyzing URLs and protect users from cyber fraud, data theft, and malicious websites. The project helps users identify suspicious websites before entering sensitive information.

Problem Statement:-

Phishing attacks are increasing rapidly, and many users unknowingly visit fake websites that steal personal information such as passwords, banking details, and login credentials. Therefore, a system is required to identify phishing URLs before users access them.

Research Findings:-

Phishing URLs often contain suspicious characters and misleading domain names.
Many phishing websites use insecure connections instead of secure HTTPS connections.
Long and unusual URLs are commonly used to trick users.
URL analysis is a simple and effective method for detecting phishing attempts.
Early detection can help prevent cyber fraud and data theft.

Methodology:-

A sample URL is selected.
The system analyzes the URL structure.
Security checks are performed on the URL.
The URL is classified as Safe or Phishing.
The result is displayed to the user.

Tools Used:-

Python
PyDroid 3
GitHub

Implementation:-

```
url = "http://fake-bank-login-security-update-example.com"

score = 0

if len(url) > 50:
    score += 1

if "https://" not in url:
    score += 1
```

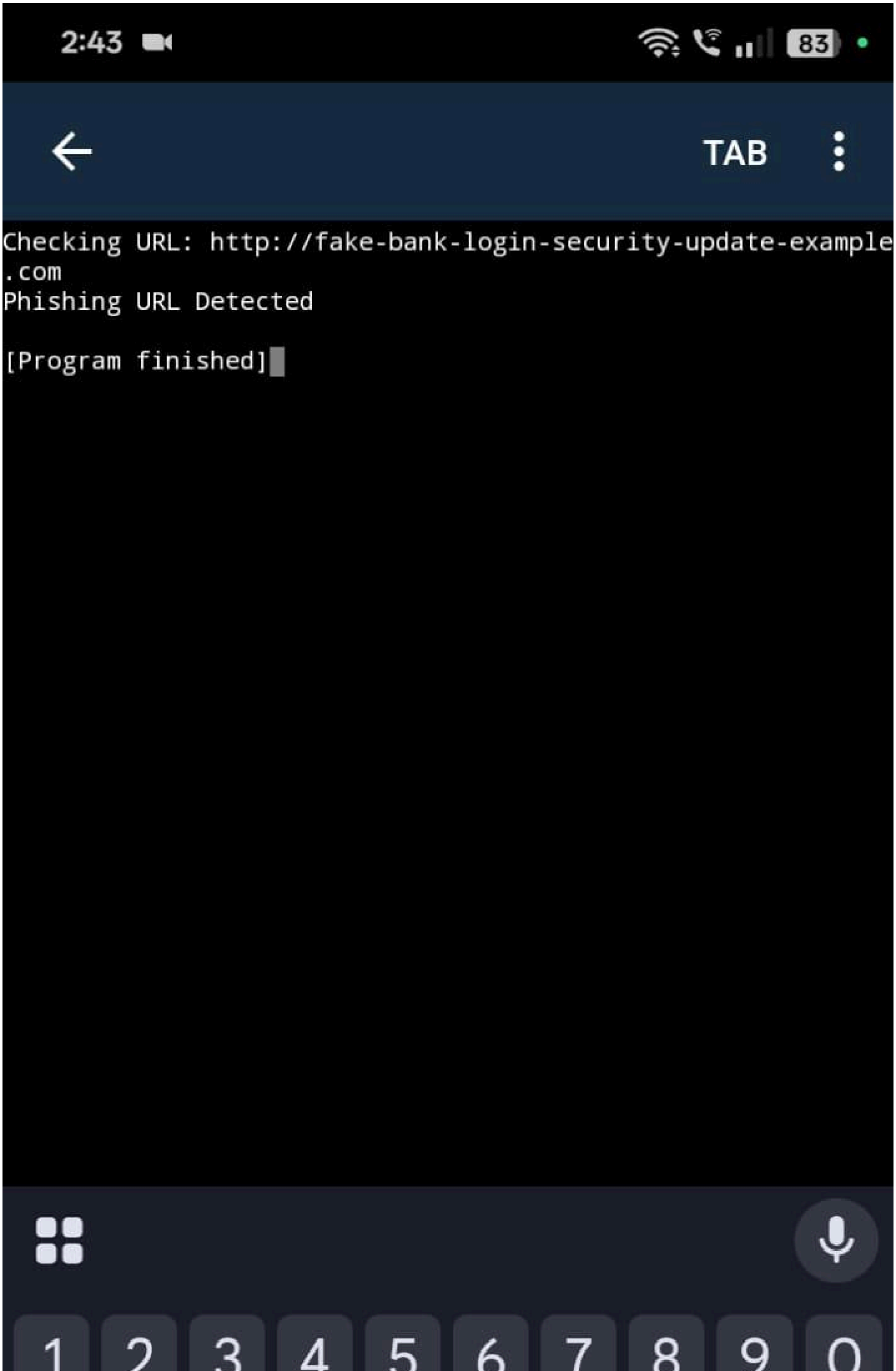
```
if "@" in url or "-" in url:
    score += 1

print("Checking URL:", url)

if score >= 2:
    print("Phishing URL Detected")
else:
    print("Safe URL")
```

<u>Screenshots:-</u>

Home Screen

URL Analysis Screen

Result Screen


2:43
83
TAB
Checking URL: http://fake-bank-login-security-update-example
.com
Phishing URL Detected
[Program finished]
1 2 3 4 5 6 7 8 9 0

Results::

Successfully analyzed the sample URL.
Detected suspicious URL characteristics.
Classified URLs as Safe or Phishing based on predefined rules.
Demonstrated basic phishing detection functionality.
Improved awareness of online security threats.

Challenges:-

Detecting newly created phishing websites.
Improving detection accuracy.
Reducing false alerts.
Handling complex URL structures.

Future Scope:-

Machine Learning Integration.
Browser Extension Development.
Real-Time URL Monitoring.
Database of known phishing websites.
Advanced URL analysis techniques.

Conclusion:-

The Phishing URL Detection System is a cybersecurity project that helps users identify malicious websites before sharing sensitive information. By analyzing URL characteristics, the system can detect potentially harmful websites and improve online security. It provides a simple, practical, and effective approach to reducing phishing attacks and protecting users from cyber threats.